

INSTRUCTION MANUAL

FOR

RESISTANCE BURB

 CMR KOREA CO., LTD.

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1. Outline

In general, the electric resistance of the metal is variable in proportion to the temperature. So, it is possible to tell the temperature by measuring the resistance, after the investigation of the relation between the electric resistance and the temperature. There is detector used the theory of the resistance variation in proportion to the temperature of the platinum wire in this result, and this detector is called PLATINUM RESISTANCE BULB.

The resistance bulb consists of the resistance element (the resistance wire wound to the bobbin) the internal leading wire, the protecting tube, the terminals, etc..

This resistance bulb has the high pure platinum wire as the resistance element which is put in the steel tube in order to protect against the circumstance condition such as the moisture, the vibration and the shock. Moreover, it has the terminals for the connection of the wire and steel equipment for fixing the part of detecting the temperature.

This resistance bulb is produced so that it can demonstrate the good characteristics for the marine use, in order to be operated under the severe vibration and the shock on board

2. Constitution

There are the resistance element and the internal leading wire inside of the resistance bulb, and the outside consists of the protecting tube, the bolts, the plugs, the terminal box, and etc.

The protecting tube - This tube is used for protection so that the resistance element and the internal leading wire may not touch the measured thing directly.

The bolts & plugs - There are the steel equipments for fixing the resistance bulb to the measuring spot.

The terminal box - This is the box to connect the resistance bulb to the outer leading wire.

3. Installation and wiring

Please pay your attention for the followings in case of the installation

- (1) It is necessary to choose the suitable place and the method for the installation which can transmit the temperature correctly and can follow the variation of the temperature as quickly as possible. Please take care not to bend the tube in view of the construction of the resistance bulb.
- (2) Please choose the suitable place which has the smallest vibration, the least dirt, and the lowest moisture. Then, please install the resistance bulb at the place which is convenient for the maintenance and the inspection.
- (3) In case of the connection of wires, please confirm the wire sign in order to avoid the mis-connection. On the treatment of the end of the wires, please connect the wires firmly by using the compression terminals to avoid the trouble such as the breaking-wire and the short-circuit.

4. Maintenance and inspection

4-1. Maintenance items and maintenance intervals

Maintenance / inspection	Initial interval	Normal interval
* Mechanical inspection	1 month	3 month
* Electrical inspection	1 month	3 month
* Cleaning	1 month	1 year

* Refer to the item 4 and 6.

4-2. The condition for opening

- (1) Turn off the switch of the electric source.
- (2) Do taping around the end of the outer lead-wire which is taken off, in order to avoid the mis-connection.
- (3) Never open the internal resistance element at the spot.

4-3. The order of opening

- (1) Take off the cover of the terminal box.
- (2) Loosen the terminal screw of the cables.
- (3) Loosen the cable gland.

4-4. The necessary tools

- (1) A plus-driver.
- (2) A spanner.
- (3) A monkey spanner (for the cable gland)
- (4) A radio pliers.
- (5) A circuit tester.

4-5. The condition for the maintenance

Prepare the spare parts.

4-6. How to keep good condition for the adjustment

(1) Mechanical inspection

Inspection points	Inspection method	Caution
a. Inspection of the looseness of the bolts, plugs and the outer tube.	Examine whether there is any looseness about the tightened part of the bolts, plugs and the outer tube.	
b. Inspection of the looseness of the terminals.	Examine whether all the terminals are fixed firmly or not.	
c. Cable gland	Examine whether there is any looseness of the lead-in cable gland.	
d. Temperature trouble	Examine whether there is any breaking of the gland cover and the wiring.	

(2) Electric inspection

Inspection of running	Inspection method	Caution
a. Inspection of running of the electric source	Examine running of the electric source with a tester.	Whether the breaking wire is or not.
b. Insulation resistance test	Examine the resistance between the terminals and the protection tube with DC 500V tester (more than 5M Ω).	

(3) Cleaning

Inspection points	Inspection method	Caution
Taking away the dirt attached with the protection tube for the high temperature.	After taking off the resistance bulb, take away the dirt attached with the insert part.	Avoidance of the error generation and the burning out.

4-7 Presumption cause or investigation method
about the trouble state

Trouble state	Inspection method	Caution
Breaking out of the wire	Thermal vibration and shock	Exchange the bad one for the spare part. The pointer of the tester does not move in case of breaking the wire and it shows less than 1MΩ in case of the bad insulation.
Bad insulation	a. As for the low temperature, the infiltration of the sea water, fresh water and the oil, due to the badness of the instruments. b. As for the high temperature the chemical change of the insulation, due to the high temperature.	
Short-circuit	Badness of the distance of the leading-wire.	
In case that the allowance increases gradually while using.	a. Badness of the indicator. b. Badness of the resistance bulb (1) The increase of the internal resistance (2) The decrease of the internal resistance (3) The increase of the dirt attached to the protection tube.	Connect the standard resistor instead of the resistance bulb, and find which of the measuring device or the detector is bad. Exchange for the spare parts. Take off the protection tube and take away the dirt.
In case that the allowance increases suddenly while using.	a. Abnormality of the measuring device b. Badness of the resistance bulb c. Bad connection of the wire, due to the looseness of the screw.	Exchange for the spare parts.

4-8 Method of repairing

- (1) Do in accordance with the reverse order of that of opening.
- (2) Concerning the one for the high temperature, the protection against burning out should be carried out around the installation screws at the side of the engine.

4-9 Confirmation items

- (1) Confirm whether the leading-wire is connected correctly.
- (2) Confirm the tightness of the cover.
- (3) Confirm the fixed part of the bolts, plugs, etc.
- (4) Turn on the switch for the electric source.

SPECIFICATION FOR THERMO RESISTANCE BULB

1. Scope

This specification applies to the thermo-resistance bulbs which used to every part of machinery.

2. Specification

- 1) Element wire : platinum (Pt)
- 2) Nominal resistance value : Pt 100 Ω at 0°C
- 3) Rated current : DC 10mA
- 4) Class : B (DIN 43760)
- 5) Resistance value of standard element : according to DIN 43760
- 6) Allowance against temperature : $\pm(0.3+0.005t)^{\circ}\text{C}$
- 7) Insulation resistance : it is to be the following table.

Temperature range(°C)	Insulation resistance (M Ω)	
	Normal temp.	Maximum temp.
-200 ~ 100	5	5
0 ~ 200	5	5
0 ~ 600	5	5

- 8) Dielectric strength : AC 500V 1 minute between conductive part to protection tube.

3. Environmental durability

Vibrational durability

It is based on the ship registers of every nation.

Approved certificates : KR, NK, LR, BV, DNV, GL.

Specification of Type MPT100

- | | |
|--------------------------|------------------|
| 1. Nominal resistance | : Pt 100 ohm |
| 2. Class | : DIN 43760 0.5 |
| 3. Regulated current | : 10mA |
| 4. Operating temperature | : 0 ~ 200°C |
| 5. Wiring | : 3-wires system |

M	P	T	1	0	0
			PT100 ohm		
			Measuring		

This MPT100 Resistance Bulb is made up of internal element, protecting tube, and terminal box, and it has the advantage of following items at fitting state.

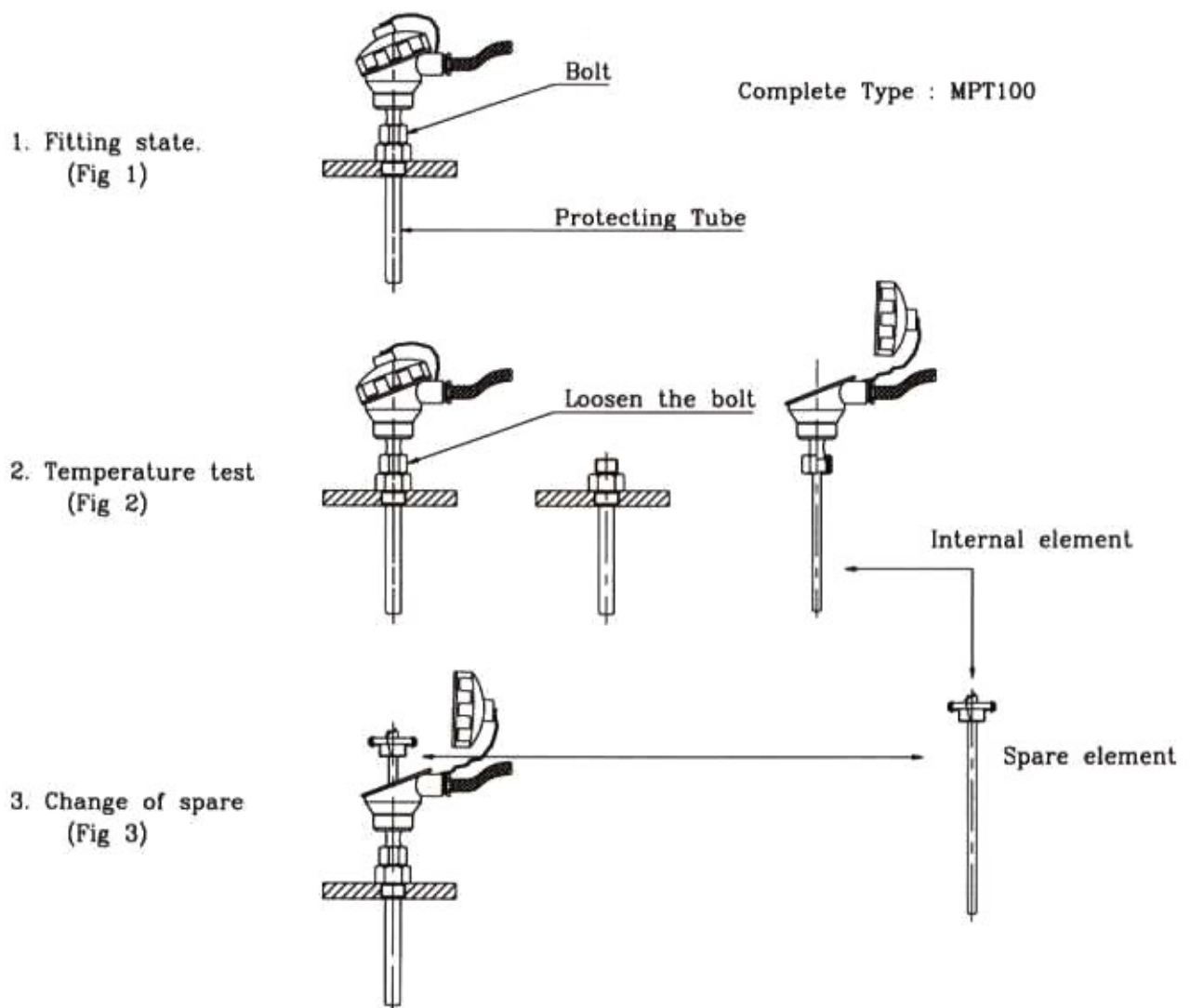
Item(1)

It can do temperature test easily without disconnection of cable from terminal box at fitting place.

Loosen the nut like as following diagram, then, internal element can be pulled out from outer protecting tube.

Item(2)

The element can be changed to new spare one at fitting place, at this time, take off the cable from terminal box, and loosen the fitting screw of terminal cover from terminal box, then element can be changed to spare one.



- CONSTRUCTION -

The temperature therminal resistance bulb is composed of a resistor element, inner wires, production tube and terminal box.

1. The resistor element is classified into two types; one for low temperature use and the other for high temperature use.

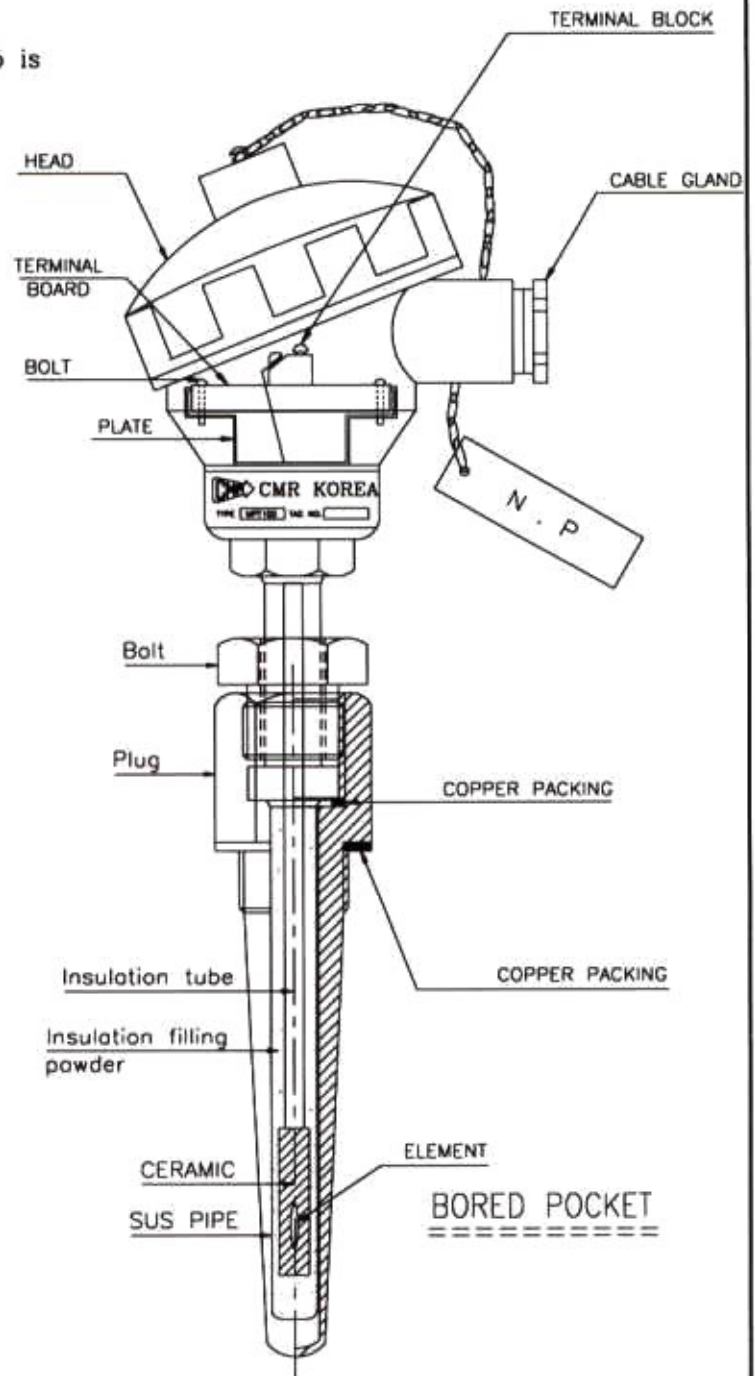
The element spool for both types are made of oxidized alumina around which a platinum wire and inner wires are wound and welded for security.

The element spool for low temperature use is provided with an insulation tube at the inner wires which are welded to the terminals through the terminal board.

The insulation tube and the terminal board use high quality ceramic as insulation material, while the terminals use selected material which is excellent instability against oxidizing and reducing atmospheres.

The element spool for high temperature is enclosed in a sleeve type insulation tube (super-ceramic), thus it is fully protected against vibrations and shocks which may occur while in use.

2. The inner wires use nickel as material and welded to the resistor element.
3. The material of the protection tube is based on the standard of SUS 304.
4. The terminal box is made up from aluminal die-cast casing.
the wire pads-through fittings are aluminal die-cast casing.
the wire pass-through fittings are based on the standard of JIS F8801, 20a.



Construction for
High Temperature Use

TABLE 1
RELATIONS TEMPERATURE/VALUE OHM

°C	-0	-1	-2	-3	-4	-5	-6	-7	-8	-9
-200	18,49	—	—	—	—	—	—	—	—	—
-190	22,80	22,37	21,94	21,51	21,08	20,65	20,22	19,79	19,36	18,93
-180	27,08	26,65	26,23	25,80	25,37	24,94	24,52	24,09	23,66	23,23
-170	31,32	30,90	30,47	30,05	29,63	29,20	28,78	28,35	27,93	27,50
-160	35,53	35,11	34,69	34,27	33,85	33,43	33,01	32,59	32,16	31,74
-150	39,71	39,30	38,88	38,46	38,04	37,63	37,21	36,79	36,37	35,95
-140	43,87	43,45	43,04	42,63	42,21	41,79	41,38	40,96	40,55	40,13
-130	48,00	47,59	47,18	46,76	46,35	45,94	45,52	45,11	44,70	44,28
-120	52,11	51,70	51,29	50,88	50,47	50,06	49,64	49,23	48,82	48,41
-110	56,19	55,78	55,38	54,97	54,56	54,15	53,74	53,33	52,92	52,52
-100	60,25	59,85	59,44	59,04	58,63	58,22	57,82	57,41	57,00	56,60
-90	64,30	63,90	63,49	63,09	62,68	62,28	61,87	61,47	61,06	60,66
-80	68,33	67,92	67,52	67,12	66,72	66,31	65,91	65,51	65,11	64,70
-70	72,33	71,93	71,53	71,13	70,73	70,33	69,93	69,53	69,13	68,73
-60	76,33	75,93	75,53	75,13	74,73	74,33	73,93	73,53	73,13	72,73
-50	80,31	79,91	79,51	79,11	78,72	78,32	77,92	77,52	77,13	76,73
-40	84,27	83,88	83,48	83,08	82,69	82,29	81,89	81,50	81,10	80,70
-30	88,22	87,83	87,43	87,04	86,64	86,25	85,85	85,46	85,06	84,67
-20	92,16	91,77	91,37	90,98	90,59	90,19	89,80	89,40	89,01	88,62
-10	96,09	95,69	95,30	94,91	94,52	94,12	93,73	93,34	92,95	92,55
0	100,00	99,61	99,22	98,83	98,44	98,04	97,65	97,26	96,87	96,48

°C	0	1	2	3	4	5	6	7	8	9
0	100,00	100,39	100,78	101,17	101,56	101,95	102,34	102,73	103,12	103,51
10	103,90	104,29	104,68	105,07	105,46	105,85	106,24	106,63	107,02	107,40
20	107,79	108,18	108,57	108,96	109,35	109,73	110,12	110,51	110,90	111,28
30	111,67	112,06	112,45	112,83	113,22	113,61	113,99	114,38	114,77	115,15
40	115,54	115,93	116,31	116,70	117,08	117,47	117,85	118,24	118,62	119,01
50	119,40	119,78	120,16	120,55	120,93	121,32	121,70	122,09	122,47	122,86
60	123,24	123,62	124,01	124,39	124,77	125,16	125,54	125,92	126,31	126,69
70	127,07	127,45	127,84	128,22	128,60	128,98	129,37	129,75	130,13	130,51
80	130,89	131,27	131,66	132,04	132,42	132,80	133,18	133,56	133,94	134,32
90	134,70	135,08	135,46	135,84	136,22	136,60	136,98	137,36	137,74	138,12
100	138,50	138,88	139,26	139,64	140,02	140,39	140,77	141,15	141,53	141,91
110	142,29	142,66	143,04	143,42	143,80	144,17	144,55	144,93	145,31	145,68
120	146,06	146,44	146,81	147,19	147,57	147,94	148,32	148,70	149,07	149,45
130	149,82	150,20	150,57	150,95	151,33	151,70	152,08	152,45	152,83	153,20
140	153,58	153,95	154,32	154,70	155,07	155,45	155,82	156,19	156,57	156,94
150	157,31	157,69	158,06	158,43	158,81	159,18	159,55	159,93	160,30	160,67
160	161,04	161,42	161,79	162,16	162,53	162,90	163,27	163,65	164,02	164,39
170	164,76	165,13	165,50	165,87	166,24	166,61	166,98	167,35	167,72	168,09
180	168,46	168,83	169,20	169,57	169,94	170,31	170,68	171,05	171,42	171,79
190	172,16	172,53	172,90	173,26	173,63	174,00	174,37	174,74	175,10	175,47
200	175,84	176,21	176,57	176,94	177,31	177,68	178,04	178,41	178,78	179,14
210	179,51	179,88	180,24	180,61	180,97	181,34	181,71	182,07	182,44	182,80
220	183,17	183,53	183,90	184,26	184,63	184,99	185,36	185,72	186,09	186,45
230	186,82	187,18	187,54	187,91	188,27	188,63	189,00	189,36	189,72	190,09
240	190,45	190,81	191,18	191,54	191,90	192,26	192,63	192,99	193,35	193,71
250	194,07	194,44	194,80	195,16	195,52	195,88	196,24	196,60	196,96	197,33
260	197,69	198,05	198,41	198,77	199,13	199,49	199,85	200,21	200,57	200,93
270	201,29	201,65	202,01	202,36	202,72	203,08	203,44	203,80	204,16	204,52
280	204,88	205,23	205,59	205,95	206,31	206,67	207,02	207,38	207,74	208,10
290	208,45	208,81	209,17	209,52	209,88	210,24	210,59	210,95	211,31	211,66
300	212,02	212,37	212,73	213,09	213,44	213,80	214,15	214,51	214,86	215,22
310	215,57	215,93	216,28	216,64	216,99	217,35	217,70	218,05	218,41	218,76
320	219,12	219,47	219,82	220,18	220,53	220,88	221,24	221,59	221,94	222,29
330	222,65	223,00	223,35	223,70	224,06	224,41	224,76	225,11	225,46	225,81
340	226,17	226,52	226,87	227,22	227,57	227,92	228,27	228,62	228,97	229,32
350	229,67	230,02	230,37	230,72	231,07	231,42	231,77	232,12	232,47	232,82
360	233,17	233,52	233,87	234,22	234,56	234,91	235,26	235,61	235,96	236,31
370	236,65	237,00	237,35	237,70	238,04	238,39	238,74	239,09	239,43	239,78
380	240,13	240,47	240,82	241,17	241,51	241,86	242,20	242,55	242,90	243,24
390	243,59	243,93	244,28	244,62	244,97	245,31	245,66	246,00	246,35	246,69

TABLE 2
RELATIONS TEMPERATURE/VALUE OHM

°C	0	1	2	3	4	5	6	7	8	9
400	247,04	247,38	247,73	248,07	248,41	248,76	249,10	249,45	249,79	250,13
410	250,48	250,82	251,16	251,50	251,85	252,19	252,53	252,88	253,22	253,56
420	253,90	254,24	254,59	254,93	255,27	255,61	255,95	256,29	256,64	256,98
430	257,32	257,66	258,00	258,34	258,68	259,02	259,36	259,70	260,04	260,38
440	260,72	261,06	261,40	261,74	262,08	262,42	262,76	263,10	263,43	263,77
450	264,11	264,45	264,79	265,13	265,47	265,80	266,14	266,48	266,82	267,15
460	267,49	267,83	268,17	268,50	268,84	269,18	269,51	269,85	270,19	270,52
470	270,86	271,20	271,53	271,87	272,20	272,54	272,88	273,21	273,55	273,88
480	274,22	274,55	274,89	275,22	275,56	275,89	276,23	276,56	276,89	277,23
490	277,56	277,90	278,23	278,56	278,90	279,23	279,56	279,90	280,23	280,56
500	280,90	281,23	281,56	281,89	282,23	282,56	282,89	283,22	283,55	283,89
510	284,22	284,55	284,88	285,21	285,54	285,87	286,21	286,54	286,87	287,20
520	287,53	287,86	288,19	288,52	288,85	289,18	289,51	289,84	290,17	290,50
530	290,83	291,16	291,49	291,81	292,14	292,47	292,80	293,13	293,46	293,79
540	294,11	294,44	294,77	295,10	295,43	295,75	296,08	296,41	296,74	297,06
550	297,39	297,72	298,04	298,37	298,70	299,02	299,35	299,68	300,00	300,33
560	300,65	300,98	301,31	301,63	301,96	302,28	302,61	302,93	303,26	303,58
570	303,91	304,23	304,56	304,88	305,20	305,53	305,85	306,18	306,50	306,82
580	307,15	307,47	307,79	308,12	308,44	308,76	309,09	309,41	309,73	310,05
590	310,38	310,70	311,02	311,34	311,67	311,99	312,31	312,63	312,95	313,27
600	313,59	313,92	314,24	314,56	314,88	315,20	315,52	315,84	316,16	316,48
610	316,80	317,12	317,44	317,76	318,08	318,40	318,72	319,04	319,36	319,68
620	319,99	320,31	320,63	320,95	321,27	321,59	321,91	322,22	322,54	322,86
630	323,18	323,49	323,81	324,13	324,45	324,76	325,08	325,40	325,72	326,03
640	326,35	326,66	326,98	327,30	327,61	327,93	328,25	328,56	328,88	329,19
650	329,51	329,82	330,14	330,45	330,77	331,08	331,40	331,71	332,03	332,34
660	332,66	332,97	333,28	333,60	333,91	334,23	334,54	334,85	335,17	335,48
670	335,79	336,11	336,42	336,73	337,04	337,36	337,67	337,98	338,29	338,61
680	338,92	339,23	339,54	339,85	340,16	340,48	340,79	341,10	341,41	341,72
690	342,03	342,34	342,65	342,96	343,27	343,58	343,89	344,20	344,51	344,82
700	345,13	345,44	345,75	346,06	346,37	346,68	346,99	347,30	347,60	347,91
710	348,22	348,53	348,84	349,15	349,45	349,76	350,07	350,38	350,69	350,99
720	351,30	351,61	351,91	352,22	352,53	352,83	353,14	353,45	353,75	354,06
730	354,37	354,67	354,98	355,28	355,59	355,90	356,20	356,51	356,81	357,12
740	357,42	357,73	358,03	358,34	358,64	358,95	359,25	359,55	359,86	360,16
750	360,47	360,77	361,07	361,38	361,68	361,98	362,29	362,59	362,89	363,19
760	363,50	363,80	364,10	364,40	364,71	365,01	365,31	365,61	365,91	366,22
770	366,52	366,82	367,12	367,42	367,72	368,02	368,32	368,63	368,93	369,23
780	369,53	369,83	370,13	370,43	370,73	371,03	371,33	371,63	371,93	372,22
790	372,52	372,82	373,12	373,42	373,72	374,02	374,32	374,61	374,91	375,21
800	375,51	375,81	376,10	376,40	376,70	377,00	377,29	377,59	377,89	378,19
810	378,48	378,78	379,08	379,37	379,67	379,97	380,26	380,56	380,85	381,15
820	381,45	381,74	382,04	382,33	382,63	382,92	383,22	383,51	383,81	384,10
830	384,40	384,69	384,98	385,28	385,57	385,87	386,16	386,45	386,75	387,04
840	387,34	387,63	387,92	388,21	388,51	388,80	389,09	389,39	389,68	389,97
850	390,26	—	—	—	—	—	—	—	—	—